

Fazul Al Rehman

Offensive Security Researcher | Bug Bounty Hunter

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EDUCATION

BS Computer Science | Institute of Business Administration (IBA), Karachi

2021 – 2025

Final Year Project: IntelliScore, AI-Based Credit Scoring System (in collaboration with Meezan Bank)

HSSC (Pre-Engineering) | Superior College of Sciences

2019

PROFESSIONAL EXPERIENCE

Independent Security Researcher | Offensive Security & Bug Bounty

2019 – Present

Full-time bug bounty hunter since Jan 2026. Offensive security research and responsible disclosure across enterprise, e-commerce, gaming, and Web3 targets.

Samsung Pass, Critical Production Vulnerability

Critical, Bugcrowd

- Broke access control to reach internal environment secrets, including a **Jenkins CI/CD Slack** token with admin-level access to sensitive operational channels.
- Mapped live dashboards, **Terraform**, SSH key rotations and internal credentials; disclosed via **Bugcrowd** for a **\$10k bounty** on a biometric platform serving millions.

Walmart Marketplace, Payment Access-Control Bypass

Critical

- Critical access-control (IDOR) flaw in **Walmart Marketplace Seller Central**: manipulating request parameters granted unauthorized read access and could reroute seller payments, placing an estimated **\$50M** in payment flow at risk.

Large-Scale Credential Exposure via Dorking

Disclosure in progress

- Systematic search-engine dorking surfaced **20,000+ exposed private keys** and API secrets across **AWS, Stripe, Twilio, Razorpay, OpenAI, Anthropic, Shopify, Twitch, and GitHub**.
- Aggregated **GitHub** keys exposed auto-deploying access to **1,600+ private repositories**; **Shopify** secret keys affected multiple e-commerce brands (Shark Tank India–featured startups, ushopmall, **Greyson Clothiers**). Coordinating responsible disclosure with affected organizations.

IBA Karachi, University Portal Security Assessment

Responsibly disclosed

- Reported multiple criticals at a top Pakistani university (personal-records database exposure; auth bypass to super-admin) that forced a full portal rebuild, now in production.

EA Origin, Security-Question Authentication Bypass

Auth bypass

- Security-question bypass in **EA Origin**'s account-recovery flow, enabling authentication without the legitimate answers.

Dell, Invoice-PDF Authorization Bypass

Auth bypass

- Authorization bypass on **Dell** allowing retrieval of invoice PDFs through serial-number enumeration.

Web3 Security Research

Ongoing

- Auditing smart contracts through **Cantina** and **Immunefi**.

Teaching Assistant | Software Architecture & Design

Aug 2025 – Dec 2025

- Mentored students on software-architecture principles and design patterns; supported course delivery and assessment.

Mindsquared | CTO (Remote)

Jan 2024 – Jun 2024

- Built data pipelines driving profitability roadmaps across **Walmart** and **Amazon** operations; helped **50+ companies** reach profitability through automated analytics.

KEY PROJECTS

Attacking & Defending Confidential VMs

Computer Security, IBA Karachi

- Reproduced HECKLER hypervisor interrupt-injection against **AMD SEV-SNP** and **Intel TDX** TEEs on an **AMD EPYC** testbed (QEMU / QMP, **Python** harness, forensic dmesg analysis).
- Proposed **Guest_Defense** (guest-side interrupt validation; advised by Dr. Faisal Iradat); reverse-engineered **Camtasia, VEGAS** and **7-Zip** to map control flow and licensing.

SHA-256 Across Kernel, User Space & Syscalls

Operating Systems, IBA Karachi

- Implemented **SHA-256** three ways across custom Linux kernels: a **kernel module** (crypto API), a **custom syscall**, and user space over **OpenSSL**.
- Benchmarked kernel/user-space boundary-crossing overhead and analyzed the **ring 0** vs user-space attack surface.

Led a four-person team from a one-line brief to a deployed, bank-facing AI product.

- Built a **semi-supervised teacher–student distillation** pipeline over **93k records**; cut validation MSE 15.67 to **3.12** ($R^2 = 0.9971$), holding $R^2 = 0.99$ held-out at **sub-2s** inference.
- Shipped the full platform in **Python / Django (PostgreSQL, Tailwind)** with a **RAG-grounded LLM** explainability chatbot and analytics dashboard.

Nested Learning & Continual Adaptation

Neural Networks & Deep Learning, IBA Karachi

- Reproduced **Nested Learning** (2025) in **PyTorch**: nested multi-timescale optimizers, a **Continuum Memory System**, a **self-modifying meta-learner** targeting **catastrophic forgetting**.
- Benchmarked against an MLP baseline to confirm optimization behaves as multi-rate memory.

AI Résumé Generator

Introduction to Text Analytics, IBA Karachi

- Fine-tuned **Llama-2 7B** with **LoRA** on 500 résumés, **RAG-grounded** for structure and tone; beat baseline **ChatGPT** head-to-head.

Real-Time E-Commerce Analytics Pipeline

Big Data Analytics, IBA Karachi

- Streaming platform over 3–4 GB of events: **Kafka** to **Spark** into **Hadoop/HBase + Redis**, **Airflow** DAGs, **Docker**.
- Ran **fraud detection**, **customer analytics** and **demand forecasting** on one live pipeline with an observability dashboard.

CartNest, Full-Stack E-Commerce Platform

Web Application Development, IBA Karachi

- **React** SPA over **Node.js/Express** and **PostgreSQL** (Docker Compose); **buyer/seller/admin** roles on **JWT + bcrypt**, with **Multer** uploads.
- Clean routes/controllers/middleware with SQL migrations and Chart.js analytics.

Student Forum Portal, Full-Stack Community Platform

Database Systems, IBA Karachi

Stack: React, TypeScript, Node.js, SQLite

- Threaded forum on a normalized schema with **foreign-key constraints**, indexed queries, **parameterized queries** (SQLi-safe) and **transactions**.

Parallel Matrix Convolution

Parallel & Distributed Computing, IBA Karachi

- Parallelized 2D convolution in **C++ / OpenMP** (four quadrants, 8 threads) to cut runtime over the serial baseline.

NayaPay Wallet Redesign

Human-Computer Interaction, Figma

- **250** high-fidelity **Figma** screens on an 11-component design system across payment, billing and onboarding.
- Full **Urdu** RTL build (124 screens) mirroring the English set, covering 12 end-to-end flows.

Autonomous Maze-Solving Robot

Arduino & Discrete Logic, 3rd of 45 teams

- Two from-scratch builds on three IR sensors: embedded C on **Arduino**, and pure combinational logic (no MCU / firmware) via **Karnaugh maps**.
- Same Boolean equations drive both builds; placed **3rd of 45 teams**.

SKILLS

Security: Bug bounty, web app pentesting, Burp Suite, sqlmap, SQLi Dumper, Nmap, reverse engineering, binary analysis, QEMU, TEEs (SEV-SNP / TDX), CTF authoring, applied cryptography

Blockchain / Web3: Solidity, Ethereum, Foundry, smart-contract auditing & PoC development, contract verification, DApps, blockchain fundamentals

Languages: Python, Java, JavaScript / TypeScript, C / C++, Assembly, SQL, Solidity, Verilog, Embedded C (Arduino)

AI / ML: PyTorch, scikit-learn, neural networks, gradient boosting, teacher–student distillation, LoRA fine-tuning, RAG, LLMs (Llama-2, Qwen 2.5, Gemini API), CUDA C/C++

Data Engineering: Apache Spark, Kafka, Airflow, Hadoop, HDFS, HBase, Hive, MongoDB, Redis, PostgreSQL, SQLite

Web & Tools: React, Node.js / Express, MERN / PERN stacks, Django, REST APIs, JWT auth, Tailwind CSS, Docker, Git, Linux, Figma

CERTIFICATIONS

- **NVIDIA Deep Learning Institute** — Getting Started with Accelerated Computing in CUDA C/C++ 2025
- **Cisco Networking Academy** — Getting Started with Cisco Packet Tracer; Exploring Networking with Cisco Packet Tracer 2025

ACHIEVEMENTS & LEADERSHIP

- **Winner, Trading Bazaar**, IBA Karachi.
- **IBA Data Science Society** — Designed and authored the cybersecurity-track **CTF** challenges and judged submissions (validating solutions, scoring participants).
- **CryptoHack** — Completed the Introduction, General, Symmetric Ciphers, Diffie–Hellman, and Hash Functions categories.